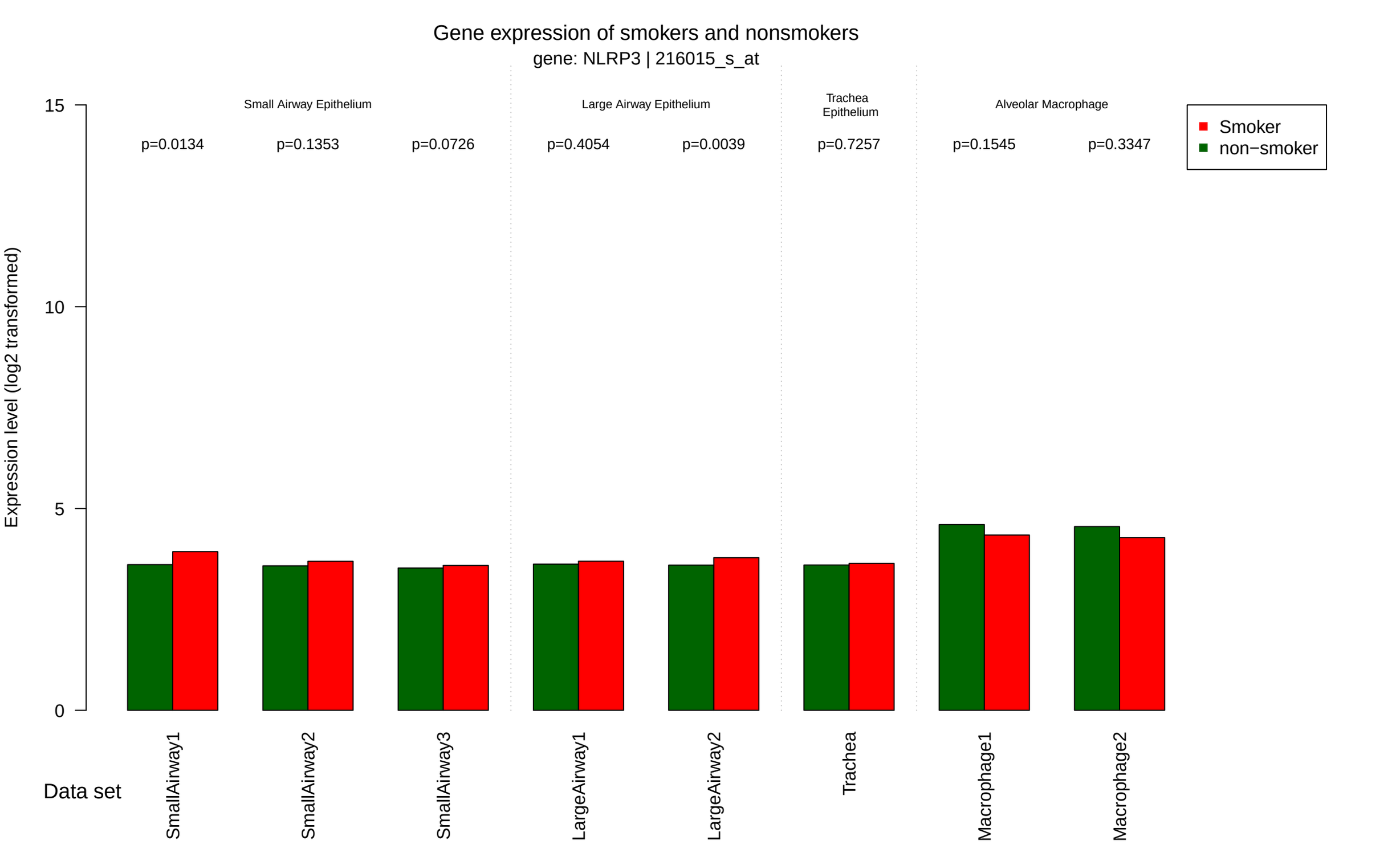
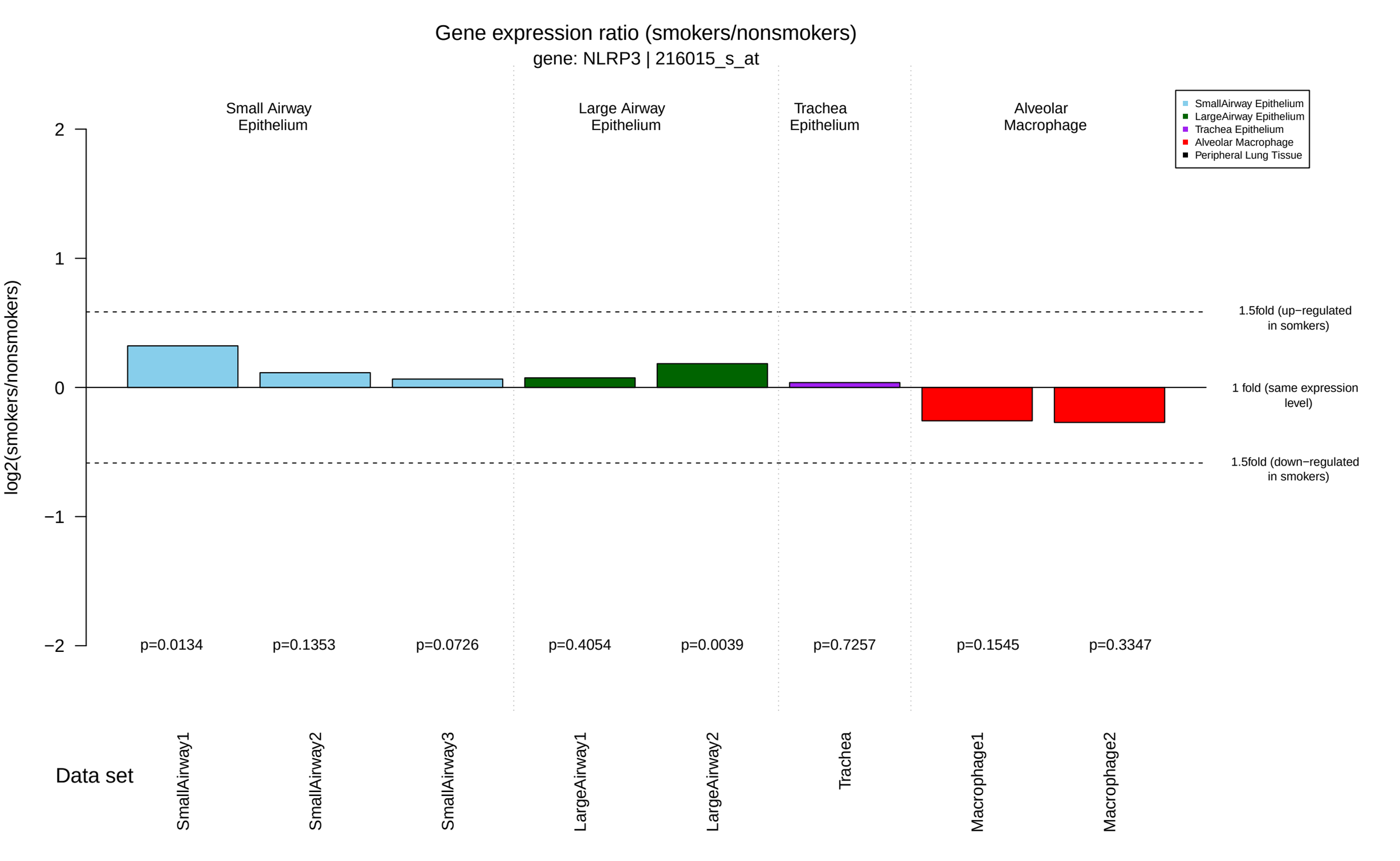
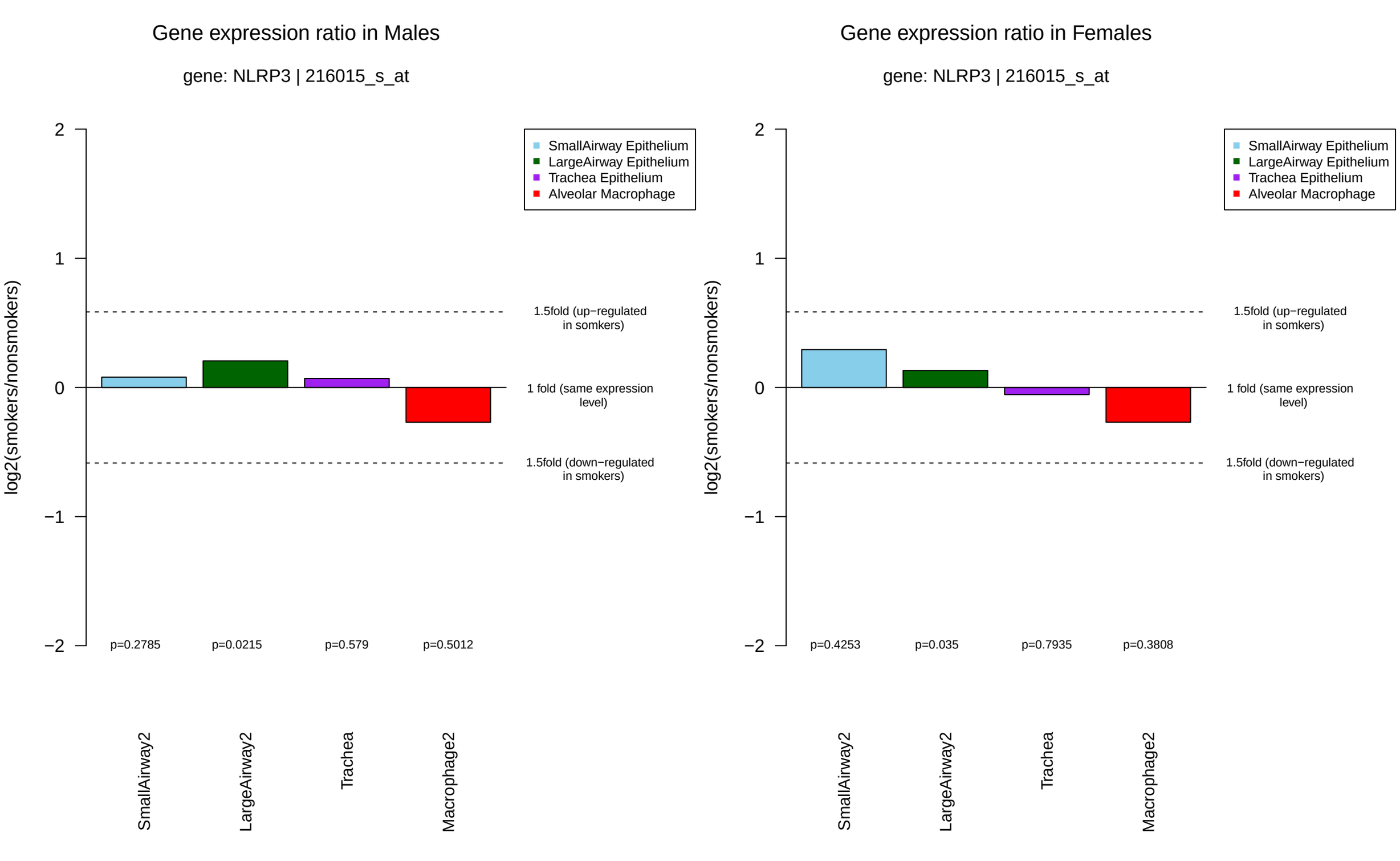
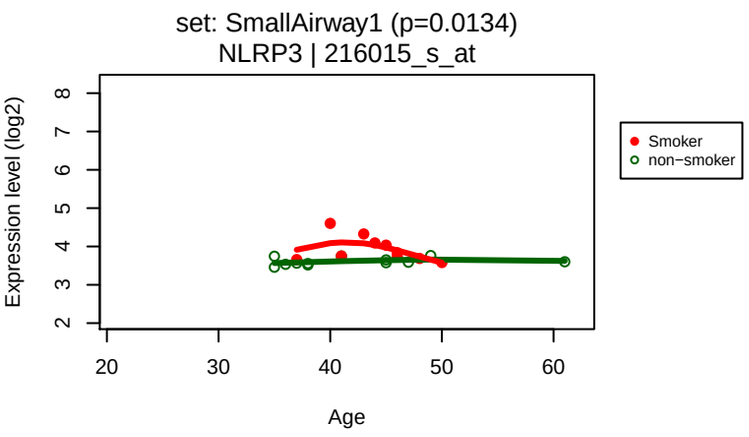
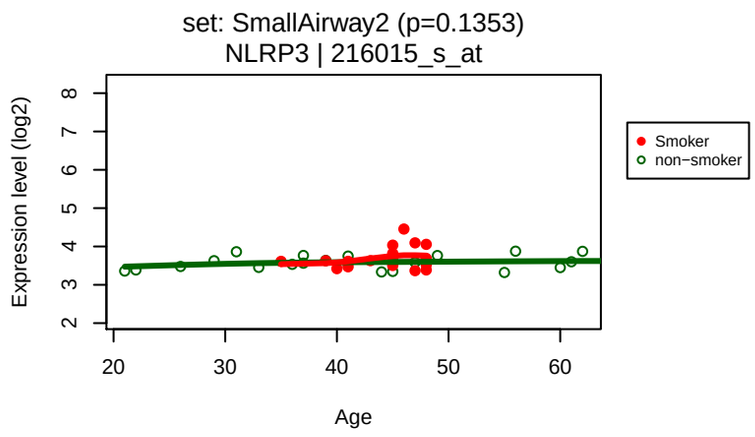
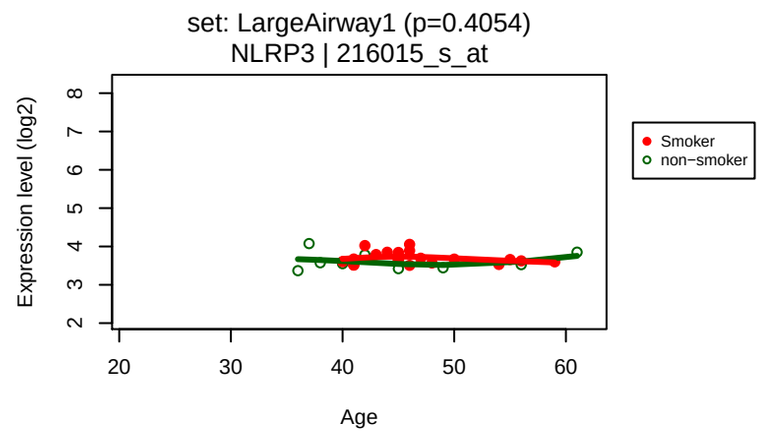

Gene expression of smokers and nonsmokers
gene: NLRP3 | 216015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0134
p=0.1353
p=0.0726
p=0.4054
p=0.0039
p=0.7257
p=0.1545
p=0.3347
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

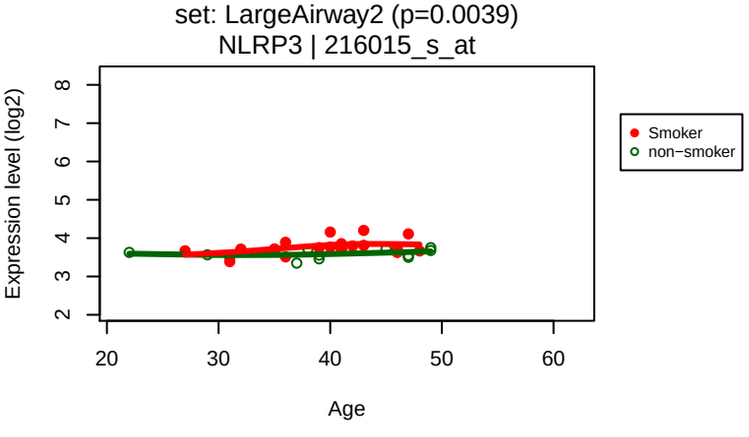

Gene expression ratio (smokers/nonsmokers)
gene: NLRP3 | 216015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0134
p=0.1353
p=0.0726
p=0.4054
p=0.0039
p=0.7257
p=0.1545
p=0.3347
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

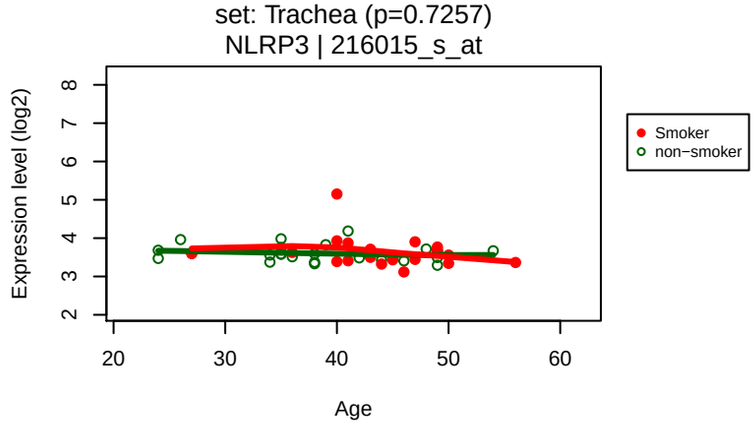

Gene expression ratio in Males
gene: NLRP3 | 216015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2785
p=0.0215
p=0.579
p=0.5012
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NLRP3 | 216015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4253
p=0.035
p=0.7935
p=0.3808
SmallAirway2
LargeAirway2
Trachea
Macrophage2

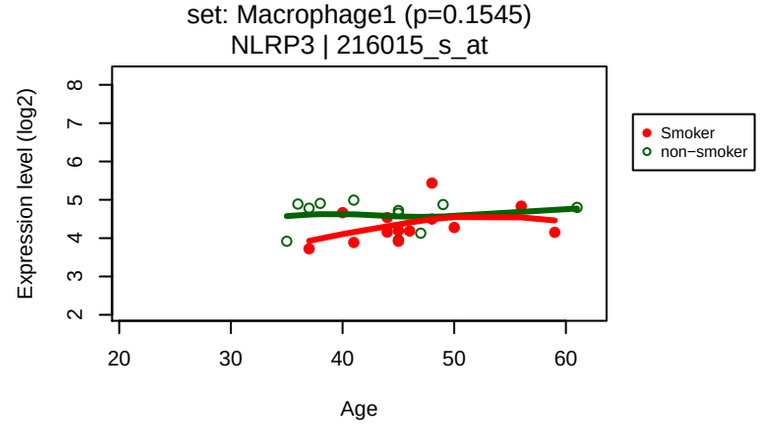

set: SmallAirway1 (p=0.0134)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker

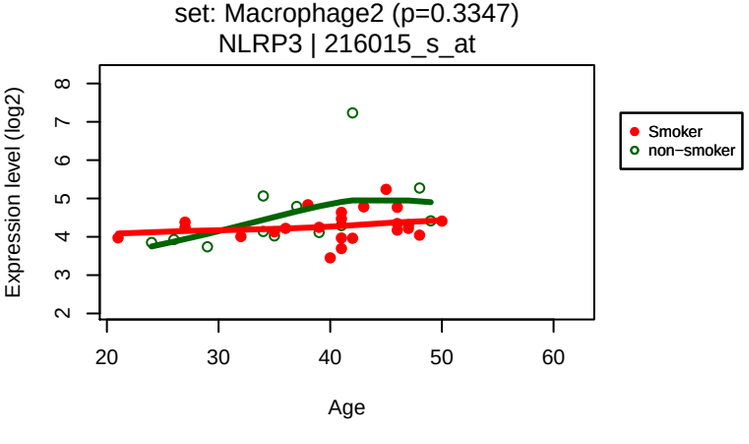

set: SmallAirway2 (p=0.1353)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker

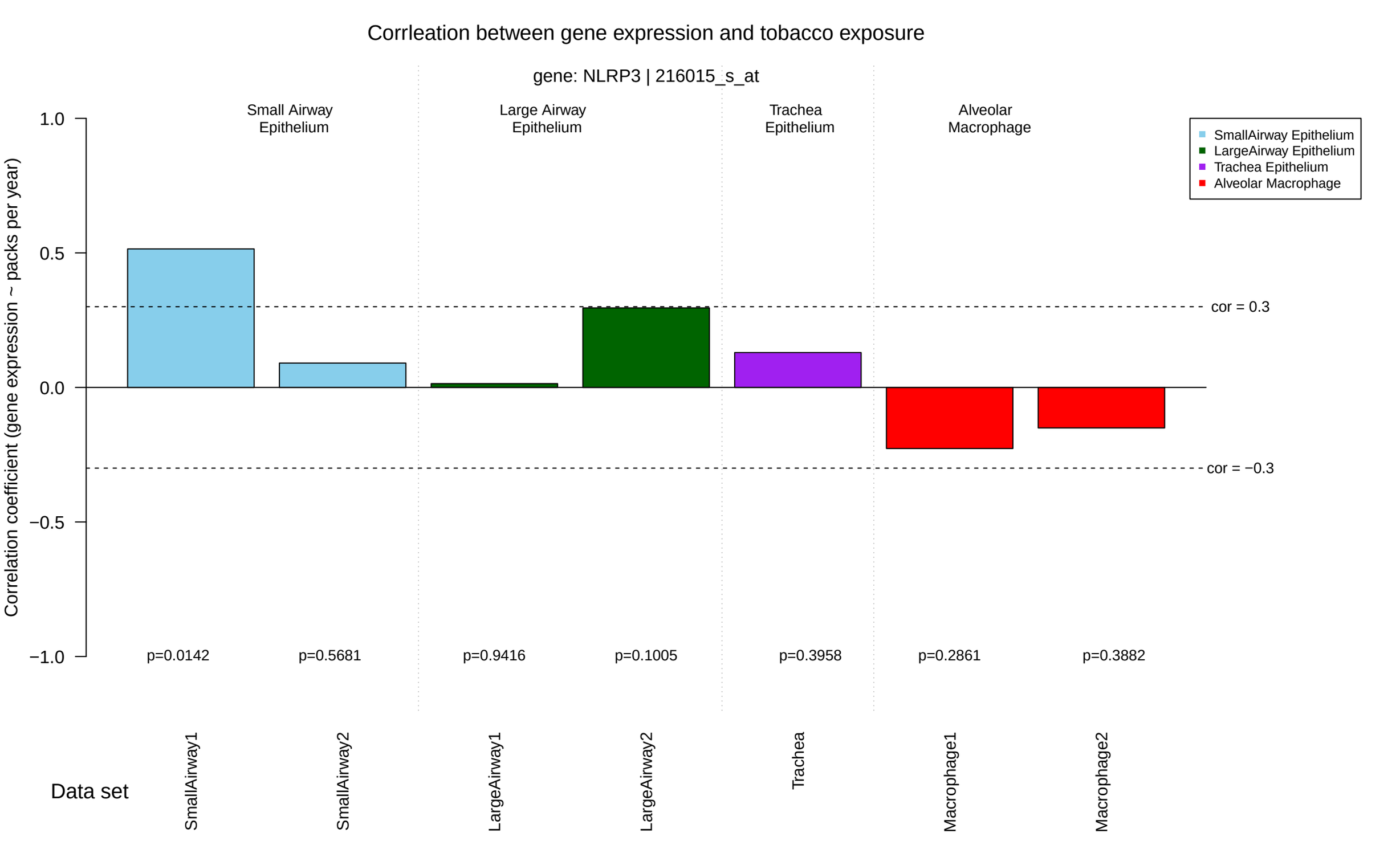

set: LargeAirway1 (p=0.4054)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0039)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7257)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1545)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3347)
NLRP3 | 216015_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NLRP3 | 216015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0142
p=0.5681
p=0.9416
p=0.1005
p=0.3958
p=0.2861
p=0.3882
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2